Cloud Attack Lab – Report

Cloud Security Attack Simulation in AWS

1. Introduction

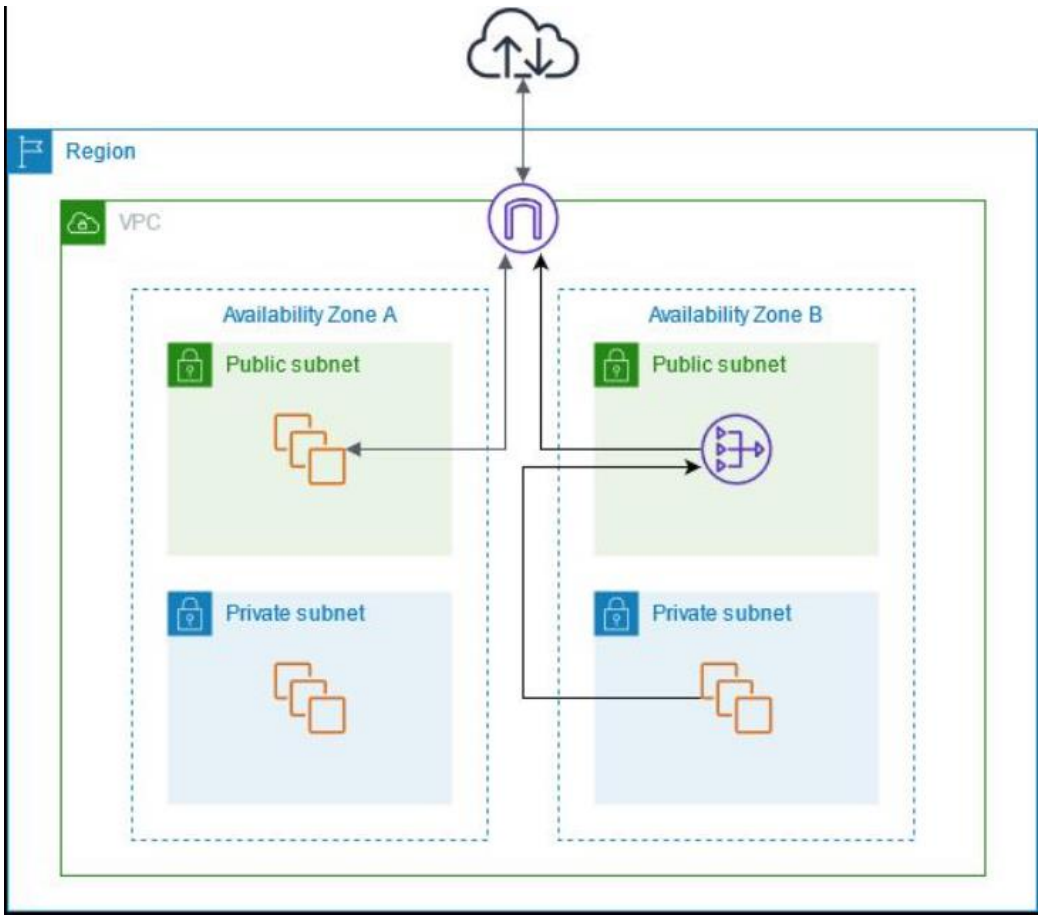
This project simulates a real-world cloud attack scenario in AWS to understand how misconfigurations in identity and access management (IAM), networking, and service exposure can lead to security risks. The objective was to analyse cloud environments from both an attacker's and defender's perspective.

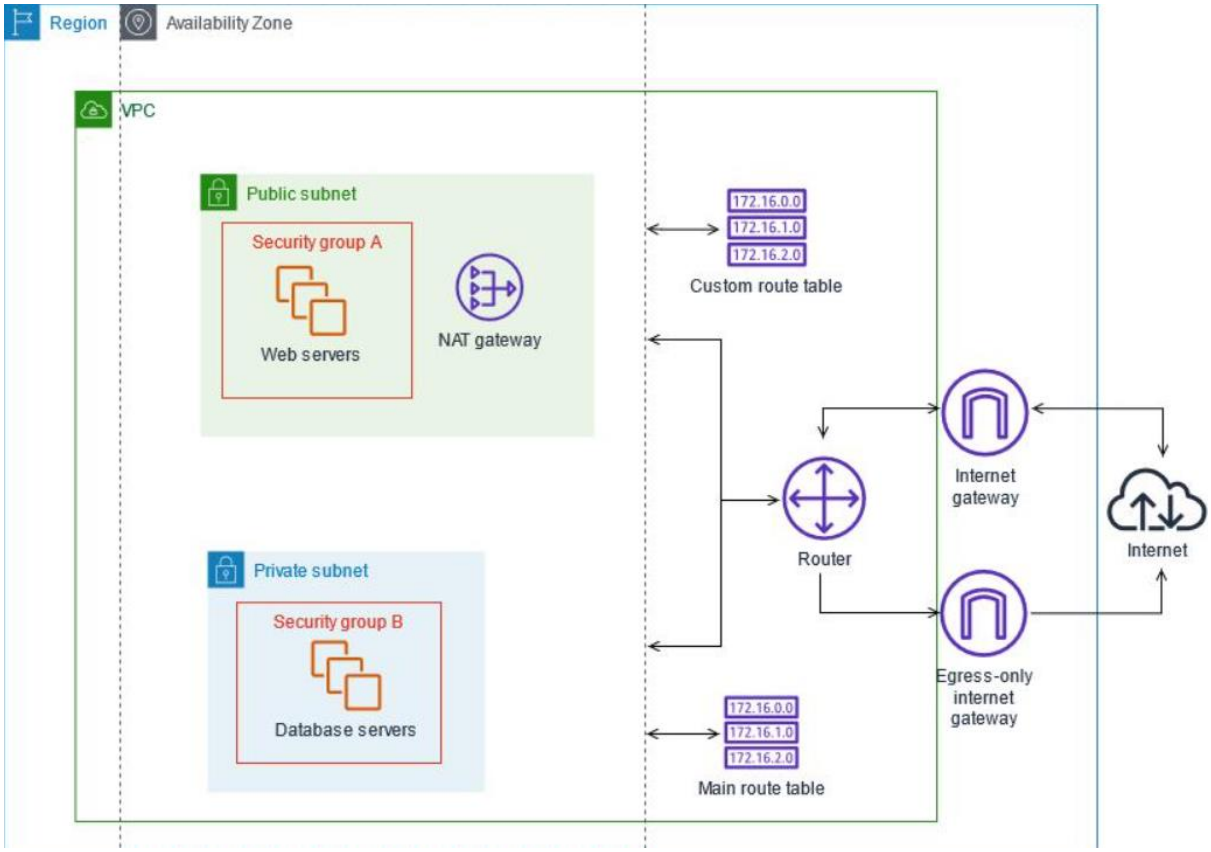
2. Architecture

The lab environment consists of:

- A public EC2 instance exposed to the internet
- Two private EC2 instances hosted in private subnets
- A Virtual Private Cloud (VPC) with subnet segmentation
- Security groups controlling inbound and outbound traffic
- IAM roles attached to instances

Architecture diagrams on the next page:


Region
VPC
Availability Zone A
Public subnet
Private subnet
Availability Zone B
Public subnet
Private subnet


Region
Availability Zone
VPC
Public subnet
Security group A
Web servers
NAT gateway
172.16.0.0
172.16.1.0
172.16.2.0
Custom route table
Internet gateway
Router
Internet
Private subnet
Security group B
Database servers
Egress-only internet gateway
172.16.0.0
172.16.1.0
172.16.2.0
Main route table

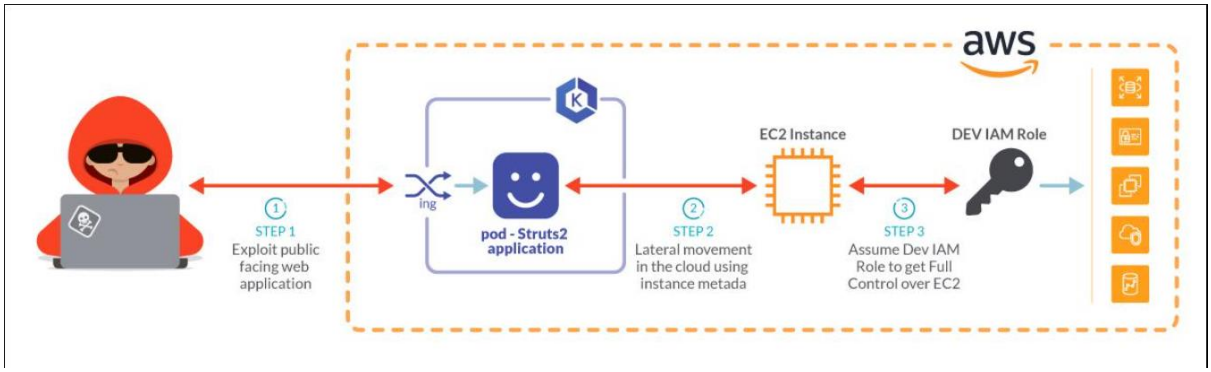
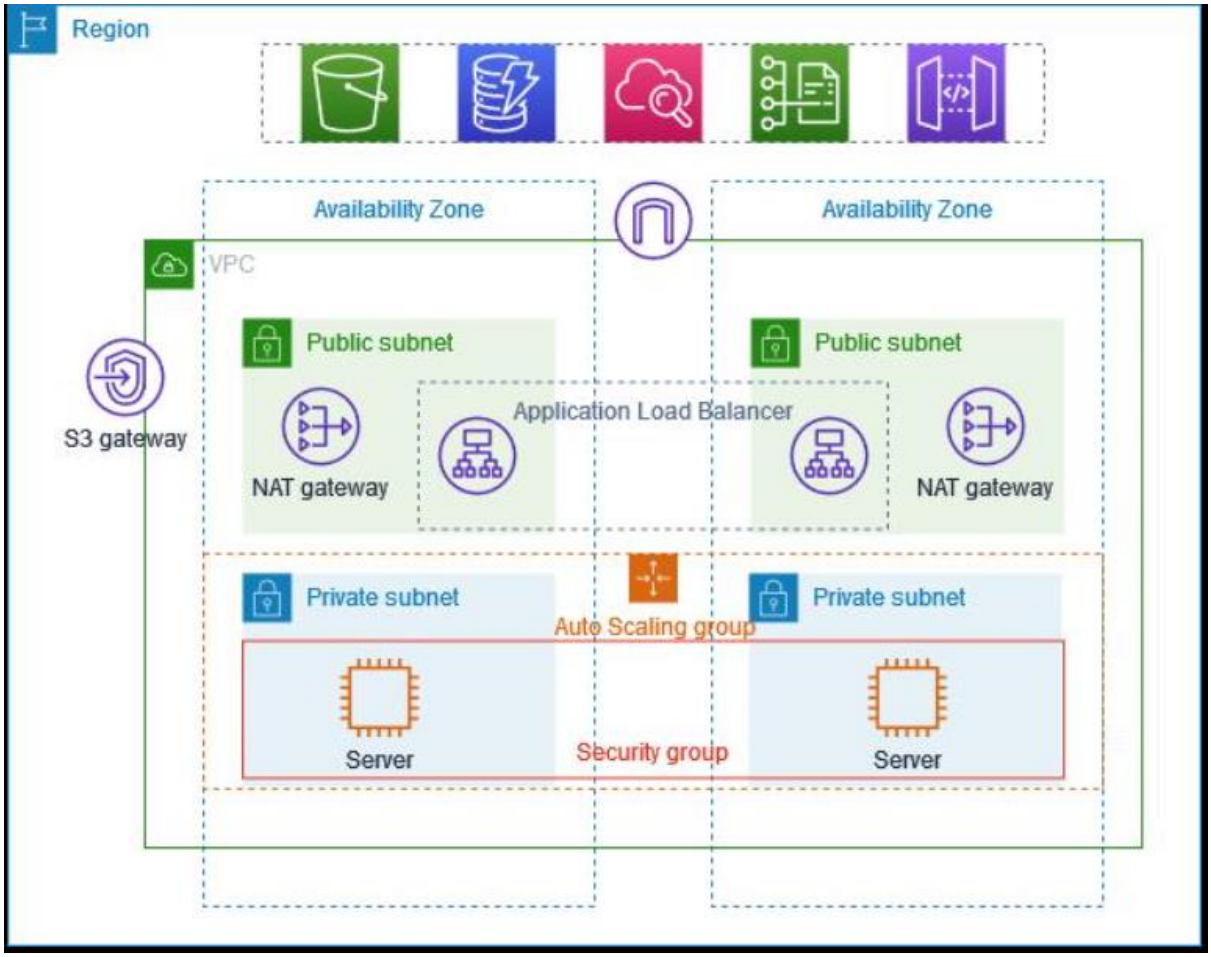
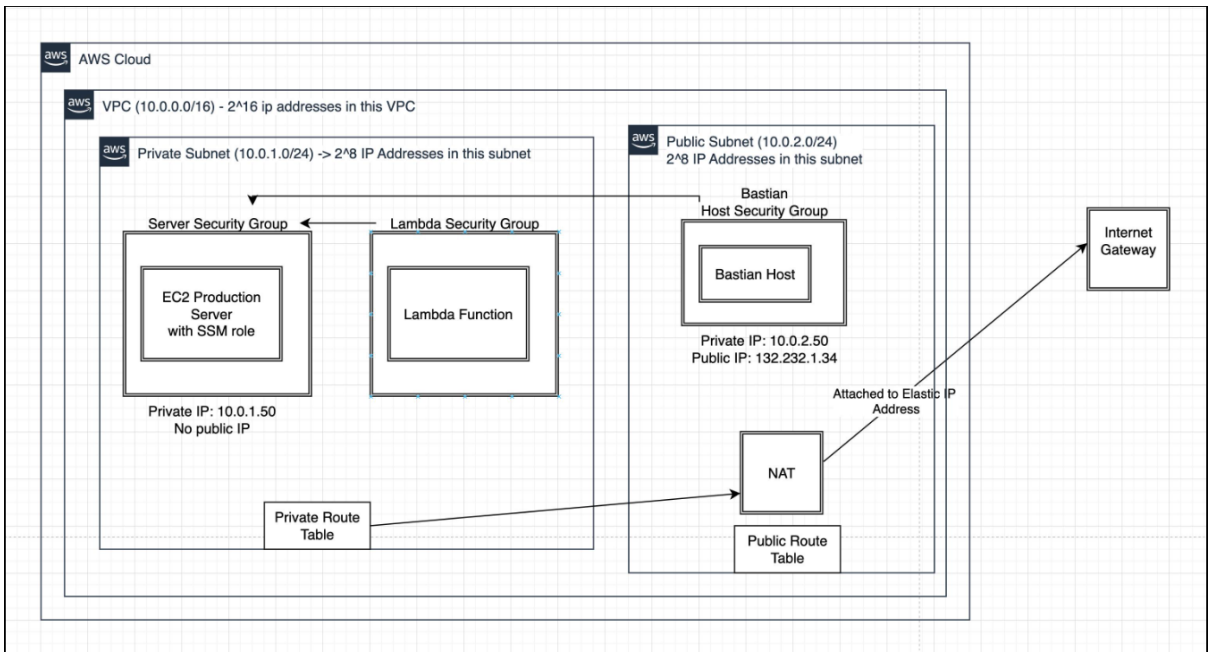


Fig: Overall flow of attack chain





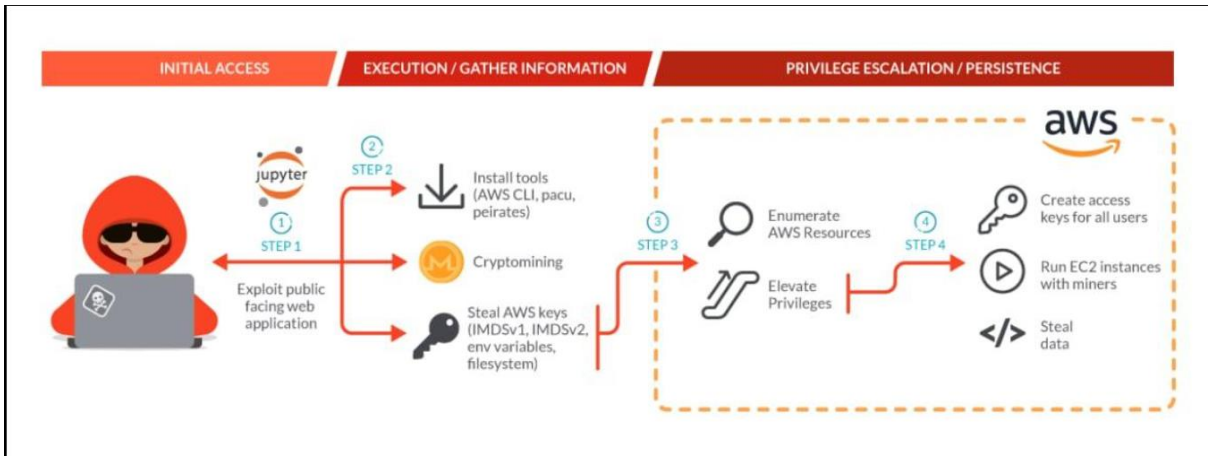


Fig: Attack flow and scenario

3. Attack Scenario

The simulation begins with an attacker gaining access to the public-facing EC2 instance. From this foothold, the following actions were performed:

- Enumeration of IAM roles using AWS CLI
- Identification of accessible resources
- Attempted lateral movement to private instances

- Testing internal service access via curl
- TLS inspection using OpenSSL

4. Tools Used

- AWS EC2, VPC, IAM
- AWS CLI
- curl
- OpenSSL
- Nmap

5. Key Findings

- Overly permissive IAM roles increase risk of privilege escalation
- Internal services can be exposed through misconfigured security groups
- Lack of strict network segmentation enables lateral movement
- Improper TLS configuration can lead to insecure communication

6. Mitigation Strategies

- **Enforce least privilege IAM policies**
- **Restrict internal traffic using security groups and subnet isolation**
- **Implement proper TLS certificate validation**
- **Enable logging using CloudTrail and VPC Flow Logs**

7. Conclusion

This project highlights how small misconfigurations in cloud environments can create significant attack paths. Understanding these risks is critical for designing secure cloud architectures and improving detection capabilities.